

torch.autograd.Function.forward#

<https://pytorch.org/docs/stable/generated/torch.autograd.Function.forward.html>

Description:

Define the forward of the custom autograd Function. This function is to be overridden by all subclasses. There are two ways to define forward: Usage 1 (Combined forward and ctx): It must accept a context ctx as the first argument, followed by any number of arguments (tensors or other types). See Combined or separate forward() and setup_context() for more details

Function Signature

```
static Function.forward(*args, **kwargs)[source]#
```

Return type

Returns:

Any

Code Examples

```
@staticmethod
def forward(ctx: Any, *args: Any, **kwargs: Any) -> Any:
    pass
```

```
@staticmethod
def forward(*args: Any, **kwargs: Any) -> Any:
    pass

@staticmethod
def setup_context(ctx: Any, inputs: Tuple[Any, ...], output:
Any) -> None:
    pass
```

Detailed Documentation

Documentation

Define the forward of the custom autograd Function.

This function is to be overridden by all subclasses. There are two ways to define forward:

Usage 1 (Combined forward and ctx):

It must accept a context `ctx` as the first argument, followed by any number of arguments (tensors or other types).

See `Combined` or `separate forward()` and `setup_context()` for more details

Usage 2 (Separate forward and ctx):

The forward no longer accepts a `ctx` argument.

Instead, you must also override the `torch.autograd.Function.setup_context()` staticmethod to handle setting up the `ctx` object. `output` is the output of the forward, `inputs` are a Tuple of inputs to the forward.

See `Extending torch.autograd` for more details

The context can be used to store arbitrary data that can be then retrieved during the backward pass. Tensors should not be stored directly on `ctx` (though this is not currently enforced for backward compatibility). Instead, tensors should be saved either with `ctx.save_for_backward()` if they are intended to be used in backward (equivalently, `vjp`) or `ctx.save_for_forward()` if they are intended to be used for in `jvp`.